What Personalised Reward Models Learn About a New Rater: Scale Use, and a Detection Limit for Content Preferences

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Abstract

Personalised reward models promise to represent how individual people judge model outputs. We test what they learn about raters they have never seen, under one protocol on three multi-rater datasets (MultiPref, DICES, PRISM): k of a new rater's ratings for adaptation, evaluation on held-out items. A text-independent scale-use baseline from measurement theory — three offsets per rater for how often they say "clearly", "slightly" or "tie" — improves held-out log-likelihood by 0.08–0.10 nats per rating at k = 20 on all three datasets. Demographic groups add nothing, a tuned low-rank user latent ties it, and reimplemented VPL and PAL and an in-context LoRA reward model gain less; on binary labels every gain is at most 0.002 nats. No method improves the prediction of which response a new rater prefers: on MultiPref and on 41,199 native pairwise choices of PRISM users, every method's gain in direction or choice accuracy over its unpersonalised version has an upper 95 % bound below one point. A semi-synthetic positive control shows what this null means. With injected content preferences that an oracle could exploit for +5 or +10 points of direction accuracy, the same methods on compressed (PCA-256) features recover less than half a point at k = 20; only at +21 points do VPL and PAL recover +6.9 and +4.6. On full features PAL detects +10 points at k = 20 and +5 points at k = 50, yet on the real data it recovers nothing (own versus another rater's ratings: +0.11 and +0.19 points). Content-level heterogeneity worth about 10 oracle points is therefore excluded on MultiPref; smaller heterogeneity remains below what 20 ratings can reveal. We recommend graded labels, new-rater evaluation with a permutation control, separate calibration and ranking metrics, and a scale-use baseline.

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1. Introduction

Reward models and LLM judges are trained on human ratings that disagree. A recent line of work treats this disagreement as signal to be modelled rather than noise to be averaged: distributional preference learning estimates a distribution over each alternative's utility rather than a point score (Siththaranjan et al., 2024); jury learning predicts individual annotators from content, annotator and demographic-group embeddings (Gordon et al., 2022), and large datasets such as PRISM are analysed by sociodemographic group (Kirk et al., 2024); personalised reward models learn a per-user latent, inferred for a new user from a few examples (Poddar et al., 2024; Chen et al., 2025). The pluralistic-alignment agenda (Sorensen et al., 2024) makes the representation of individual and group differences a first-class goal.

What these models learn about a person is rarely tested directly. Evaluations typically use synthetic users whose preferences differ by construction, binary preference labels, or users seen in training. We ask a simple question under a common protocol on real raters: **given k ratings from someone the model has never seen, how much better can it predict their other ratings — in particular, which response they prefer — and what is it that it learns?** The answer we find is that the methods learn calibration, not ranking — and a semi-synthetic positive control shows how strong a rater's content preferences would have to be before they learn ranking.

Measurement theory suggests a candidate answer that reward modelling largely ignores: raters differ in **how they use the response scale** — some say "clearly better" where others say "slightly better", some rarely call a tie, some are harsher across the board (Linacre, 1989; Baumgartner & Steenkamp, 2001). Such response styles are text-independent, cheap to estimate, and invisible to models that condition only on content or demographics. We make them a reference point for calibration. Because they are text-independent they barely change which response a rater is predicted to prefer (only through asymmetric "clearly"/"slightly" mass); a personalised reward model that

learns content preferences should therefore show its value in ranking, and a model whose likelihood gain does not exceed the scale-use offsets has not shown that it learns more than scale use.

Contributions. (1) A learner-free measurement of where the predictable structure of multi-rater preference data sits — split-half reliability of the rating distribution's shape, cross-fitted text decodability, and a rater–item decomposition of held-out likelihood (§3). (2) A three-parameter scale-use baseline that plugs into any reward model with graded outputs (§4). (3) A systematic new-rater test, on three datasets, of group-conditioned, low-rank-latent, published (VPL, PAL), in-context and fine-tuned personalised reward models against that baseline, documented in an analysis log of timestamped internal protocols, with post-result changes marked as amendments and failures reported (§5, §9). (4) Two limits of what personalisation buys — it disappears on binary labels and it is mostly calibration rather than ranking — and recommendations for the field (§6–7). (5) A semi-synthetic positive control with a permutation contrast that measures how strong content preferences must be for these methods to detect them, and places the real data on that scale (§5.7).

2. Related work

Pluralistic and distributional reward models. Distributional preference learning (Siththaranjan et al., 2024), diverging-preference detection (Zhang et al., 2025), PRISM (Kirk et al., 2024), MultiPref (Miranda et al., 2025) and DICES (Aroyo et al., 2023) motivate modelling disagreement; jury learning (Gordon et al., 2022) and multi-annotator models (Mostafazadeh Davani et al., 2022) predict individual annotators. **Personalised reward models** infer a user latent from a few examples (Poddar et al., 2024; Chen et al., 2025), represent users in a low-rank reward subspace (Bose et al., 2025), induce persona prompts from a user's history (Ryan et al., 2025, who also release a benchmark of PRISM and Chatbot Arena users), or learn per-user embeddings jointly with the reward model (Li et al., 2024); VPL's latent can collapse under sparse preference data (Kim & Kim, 2026). A multi-faceted comparison of personalisation methods (Dong et al., 2025) reports large differences between methods and safety costs. Annotator embeddings (Deng et al., 2023) and per-annotator heads (Mostafazadeh Davani et al., 2022) are close relatives of our offsets; our contribution is not the offset model but the new-rater test and what it shows. Rater groups have been found to use safety severity scales differently (Mishra et al., 2026), which is consistent with the leniency component we find on DICES. **Annotator demographics** as predictors of labels have been found weak in several studies (e.g. Orlikowski et al., 2023). **Measurement theory**: rater severity and scale use are standard facets in many-facet Rasch measurement (Linacre, 1989; Eckes, 2011) and response styles are a classical survey-methods topic (Baumgartner & Steenkamp, 2001). We connect the last line to the first: for new-rater prediction, the measurement-theory rater facet is what the reward model is missing. **Collaborative filtering and annotator models.** User and item bias terms carry much of the predictive gain in collaborative filtering (Koren et al., 2009), and user-specific ordinal thresholds predict personal rating distributions (Koren & Sill, 2011); annotator models estimate rater-specific labelling behaviour (Dawid & Skene, 1979; Raykar et al., 2010; Hovy et al., 2013), and the label-variation literature argues for modelling it (Plank, 2022; Uma et al., 2021). Our offsets are of this kind and are not new; what is new is testing personalised reward models against them on new raters, and measuring what content personalisation these models could detect at all.

3. Where is the predictable structure? A learner-free measurement

Data. MultiPref (Miranda et al., 2025; 10,461 comparisons; 4 ratings each by 2 crowd and 2 expert raters; 227 persistent raters; 5 ordered levels A-clearly … B-clearly). PRISM (Kirk et al., 2024; 68,371 responses each rated 1–100 by one of 1,396 users; binned to 5 fixed levels). DICES-350 (Aroyo et al., 2023; 350 conversations fully crossed with 104 quality-checked raters; safe / unsure / unsafe).

Estimators. For an item with ratings on K ordered levels, let h be the empirical rating distribution and C the cumulative operator. Split the item's raters at random into disjoint panels P_1 (a raters) and P_2, and let q be a cross-fitted *location-only* forecast with exactly the mode and mean of P_1 (built from other items in the same mode-and-mean cell). With $r_1 = C(h_1 - q)$, $r_2 = C(h_2 - q)$, $M = \|r_1\|^2/n$ and $X = \langle r_1, r_2\rangle/n$, the forecasting value of the raw distribution over its own location obeys the identity $S(h_1) - S(q) = M - 2X$ under the ranked probability score, so $\rho = X/M$ is a split-half reliability of the distribution's *shape* and the raw distribution beats its location exactly when $\rho > 1/2$. Text decodability is the proper-score gain $G = (2\langle \hat{r}, r_2\rangle - \|\hat{r}\|^2)/n$ of a residual $\hat{r}$ decoded from frozen text features, cross-fitted over prompt groups, reported as the fraction G/X of the reproducible shape. The rater–item decomposition fits ordered-probit models $Y^*_{ir} = \theta_i + \delta_r + \sigma_i\sigma_r\varepsilon$ with item locations θ_i, rater shifts δ_r, rater scales σ_r and item dispersions σ_i, compares held-out likelihood over random cells (every held-out rater and item appears in training), and chooses all penalties by inner cross-validation.

Reproducibility. With the split-half estimator of a companion study (anonymised), the reliability of the rating distribution's shape beyond its exact mode and mean is $\rho = 0.25$

[0.22, 0.29] on MultiPref overall preference with three raters (helpful 0.29, harmless 0.21): the raw few-rater distribution forecasts a new rater worse than its own location ($\rho < \frac{1}{2}$). **Text decodability.** A cross-fitted ridge decoder on frozen Qwen2.5-7B-Instruct features, scored on an independent rater panel, recovers G/X = 10.5 % [5.7 %, 11.9 %] of the reproducible shape (within-cell permutation control ≤ 0; positive control 0.35) — comparable to the upper end for ordinal quality ratings (upper bounds 0.4–10 %) and below NLI (17–26 %). **Rater vs item.** A long-format ordered-probit decomposition with real rater identities and penalties tuned by inner cross-validation: rater-specific thresholds improve held-out likelihood by +0.036 nats per rating over item location alone, item-specific dispersion by a further +0.0014 (ratio 25; 19–36 across the five folds); of the rater part, rater scale contributes +0.031 and rater shift +0.005.

Beyond item location, the predictable structure is therefore mostly rater-side scale use, while the item-level shape that replicates across disjoint panels is small and only partly in the text. These two facts come from different estimands and do not say that disagreement is "not in the text". The stage-1 decision rule we had written down returned **"mixed"**, because G/X was not ≤ 10 % (interval 5.7–11.9 %); the factorised model of §4 was chosen with that outcome known.

4. A three-parameter scale-use baseline

(Figure 2 illustrates the effect of the offsets.) Let $\ell(x) \in R^K$ be the class logits of a text-only distributional reward model for item x (K ordered rating levels). For rater r we add a text-independent offset vector: $p(y \mid x, r) = \mathrm{softmax}(\ell(x) + b_r)$. Variants: **DF** ($b_r \in R^K$, free), **DFs** (two-sided preference scales only: b_r tied symmetrically, (e, s, t, s, e) — extreme, slight, tie — 3 parameters, no A-vs-B position term). Rater offsets have a Gaussian prior centred on the training-rater mean; a new rater's offsets are the MAP estimate from k of their ratings with the text model frozen. Baselines under the same protocol: **D** (text-only distributional RM), **DG** (group offsets: crowd/expert, race, region), **DL** (low-rank text-interacting latent: $\ell(x) + (Wx + w_0) \cdot z_r$, d = 8, the personalised-RM family). An interpretable probit family (rater shift / log-scale on an ordered-probit text head) is reported as a secondary analysis.

Training and adaptation. The text model and the training raters' offsets are fitted jointly by penalised maximum likelihood,

$$\min \sum_{(i,r,y)} -\log p(y \mid x_i, r) + \lambda_{\text{text}} \|V\|^2 + \lambda_r \sum_r \|b_r\|^2,$$

with λ_{text} chosen on the text-only model and λ_r on held-out items of the training raters. A new rater r^* is adapted from k ratings $\{(x_j, y_j)\}$ by $\hat{b}_{r^*} = \arg\min_b \sum_j -\log \mathrm{softmax}(\ell(x_j)+b)_{y_j} + \lambda_r \|b - \bar{b}\|^2$, with $\bar{b}$ the training-rater mean (the prior mean; $\hat{b} = \bar{b}$ at $k = 0$) and the text model frozen — a convex problem in 3–5 parameters, solved in milliseconds. Any text model that outputs graded log-probabilities can be used: we use a ridge head on frozen LLM features (§5.1–5.3), a LoRA-fine-tuned LLM reward model (§5.4) and the decoder of a published personalised reward model (§5.5).

5. Experiments

Protocol. 5 folds; in each, 20 % of raters (stratified by group) and 20 % of prompt groups are held out; models are trained on training raters × training items with penalties chosen on a validation split; each test rater is adapted on $k \in \{0, 5, 10, 20, 50\}$ of their ratings of *training* items (5 random draws) and evaluated on their ratings of *held-out* items. Primary metric: held-out log-likelihood per rating; also RPS and preference-direction accuracy; rater-cluster bootstrap (B = 2,000). Because test raters share test items on MultiPref and DICES, the headline k = 20 contrasts were also re-estimated with a two-way bootstrap that resamples raters and prompt groups independently: on graded labels every log-likelihood interval widens by at most 0.007 on either side and no log-likelihood conclusion changes; direction-accuracy differences of about a point or less (the style model's +0.5 on MultiPref, VPL's direction advantage over the style model) lose significance and are marked where they appear. Text model in §5.1–5.3: frozen Qwen2.5-7B features with a ridge head; §5.4: a LoRA-fine-tuned reward model.

5.1. Main result (MultiPref)

Figure 1 shows the gain over the text-only reward model as a function of k on all three datasets. In the table, DF/DFs/DG use the training-rater prior and DL and the DFs − DL column use the few-shot-tuned prior (the DFs − DL contrast is post hoc).

Preference-direction accuracy (D: 0.674): DFs +0.005 [+0.001, +0.008] at k = 20 (not significant once items are also resampled: [−0.001, +0.011]); DF −0.014 [−0.020, −0.007]. A three-number style model gains as much as a tuned 8-dimensional text-interacting latent (tie at k = 20) and more at k = 5, 10 and 50; fitting a rater's A-vs-B position from few ratings costs direction accuracy.

Reference points. To size these gains: on MultiPref at k = 20 a uniform forecast scores −1.609 nats per rating, a forecast that ignores the text and uses only the new rater's 20 ratings −1.487, the text-only model −1.441, the text model with the three offsets −1.342, and offsets fitted in-sample on all of the rater's held-out ratings (an optimistic ceiling for text-independent offsets) −1.262. Twenty ratings of a person, without the text, recover 73 % of the text model's

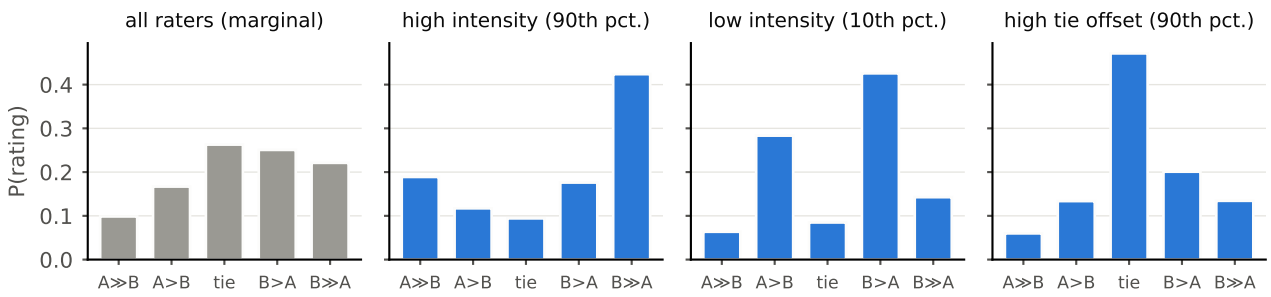


Figure 1. What three numbers per rater change. Left: the marginal distribution of MultiPref ratings. Right: the same distribution passed through the fitted scale-use offsets (extreme, slight, tie) of three real new raters (k = 50) closest to the 90th / 10th percentile of intensity (extreme minus slight) among raters whose tie offset is at or below the median, and closest to the 90th percentile of the tie offset. The offsets are text-independent: they reshape how strongly a rater is predicted to prefer, not which side.

k	DF − D	DFs − D	DG − D	DL − D (tuned prior)	DFs − DL (tuned, post hoc)
5	+0.028 [+0.012, +0.044]	+0.048 [+0.029, +0.070]	≈ 0	+0.023 [−0.002, +0.047]	+0.026 [+0.016, +0.037]
20	+0.080 [+0.055, +0.106]	**+0.100 [+0.074, +0.128]**	≈ 0	+0.095 [+0.069, +0.122]	+0.0045 [−0.0004, +0.0092]
50	+0.114 [+0.087, +0.142]	+0.128 [+0.101, +0.156]	≈ 0	+0.120	+0.007 [+0.002, +0.013]

gain over uniform; the offsets close 56 % of the gap between the text-only model and the in-sample ceiling. On DICES the rater's ratings alone predict better than the text alone (−0.766 vs −0.840 at k = 20). Even the in-sample ceiling raises MultiPref direction accuracy by only 1.3 points (0.674 → 0.687), confirming that text-independent offsets cannot carry ranking.

5.2. Replications (k = 20)

DF − D and DG − D use the training-rater prior, DF − DL the few-shot-tuned prior. Direction accuracy on PRISM means good-vs-bad around the scale midpoint.

5.3. What the offsets encode

In the probit family, rater *scale* matters more than rater *shift* on MultiPref (+0.036 [+0.020, +0.057] at k = 20), but rater *shift* dominates on DICES (−0.053) and PRISM (−0.031): leniency/strictness on single-response rating scales, intensity on two-sided preference scales.

5.4. On a fine-tuned reward model

With a LoRA-fine-tuned Qwen2.5-7B reward model (D-LoRA, trained per fold on the training raters' rating distributions) as the text model, the three-number style model still adds +0.100 [+0.071, +0.130] nats per rating at k = 20 — the same as its gain on the frozen-feature head (+0.100). Fine-tuning improves the text-only model by +0.027 [+0.016, +0.039] (direction accuracy +1.4 points) and the style model by the same +0.027: the two gains add. Personalising the frozen-feature model with 20 ratings per rater gains more than fine-tuning the text model (+0.072 [+0.041, +0.106]). DFs keeps the direction accuracy of D-LoRA (+0.002 [−0.002, +0.006]). Against in-context personalisation — a separate LoRA model on the same base, trained with ten of the rater's other ratings in the prompt and evaluated with the new rater's k ratings in the prompt — the three offsets on D-LoRA are better at k = 20. The in-context model's result varies with its training seed (two of fifteen runs failed, gaining −0.078 and −0.055 over D-LoRA against +0.021 to +0.125 for the others; the two failed runs have the highest training loss), so it was trained with three seeds per fold, and we fixed in advance that "better" requires a lower bound above zero for every seed and for the seed average. Averaged over seeds — the expected result of one training run — the offsets are better by +0.041 [+0.027, +0.054] nats per rating (209 raters; per seed +0.062, +0.018 and +0.042, every lower bound at least +0.005) and more accurate on which response the rater prefers (+1.9 points [+0.9, +2.7]). The in-context model does learn from the history (+0.051 [+0.024, +0.079] over D-LoRA on average), at several thousand prompt tokens per prediction, but its direction accuracy is below D-LoRA's (−1.7 points [−2.6, −0.8] on average; −0.6 to −3.0 per seed).

5.5. Published personalised reward models

We reimplemented VPL (Poddar et al., 2024) and PAL-B (Chen et al., 2025) from their official code as heads on the same frozen per-response embeddings and evaluated them under the same new-rater protocol, refitting the text-only and style models on those embeddings. On graded

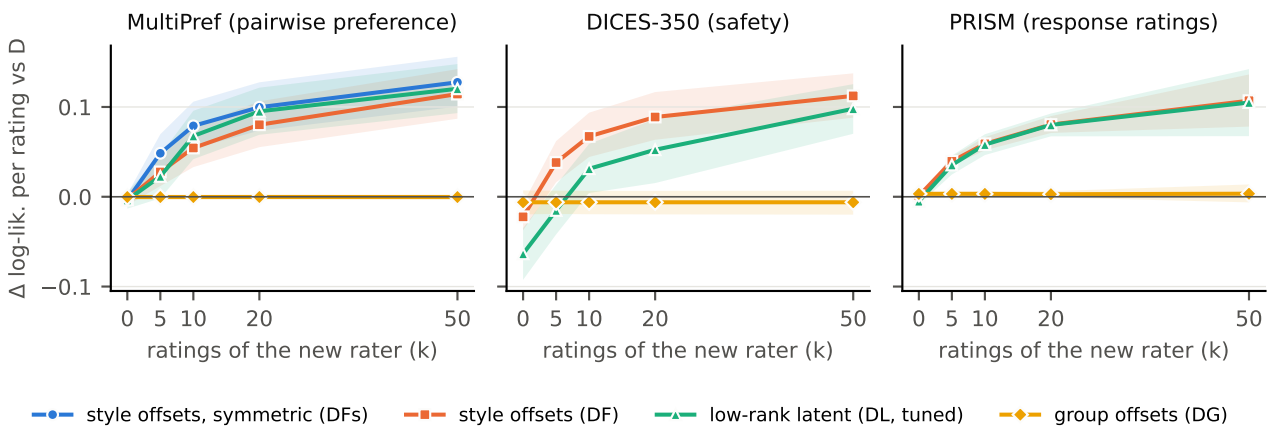


Figure 2. New raters: gain in held-out log-likelihood per rating over the text-only distributional reward model (D), as a function of the number k of the new rater's ratings used for adaptation; 95% rater-cluster bootstrap bands, 5 folds. Text-independent rater offsets (DF; symmetric three-parameter DFs on two-sided preferences) gain 0.08–0.10 nats per rating at k = 20 on all three datasets; group offsets (crowd/expert, race, region) gain nothing; a low-rank text-interacting rater latent with a few-shot-tuned prior ties the three-parameter DFs and beats the five-offset DF on MultiPref (DF − DL = −0.015). DF, DFs and DG use the training-rater prior; DL the few-shot-tuned prior.

dataset	DF − D	DG − D (groups)	DF − DL (tuned prior)
DICES-350 (safety)	+0.089 [+0.064, +0.117]	−0.006 (race)	+0.037 [+0.022, +0.053]
PRISM (ratings)	+0.080 [+0.071, +0.090]	+0.003 (region)	−0.001 [−0.006, +0.003] (formally not non-inferior: bound −0.006 < −0.005)

labels, at k = 20 the style model is better than VPL by +0.092 [+0.068, +0.119] nats per rating ([+0.063, +0.124] when prompt groups are also resampled) and than PAL-B by +0.158. VPL's latent does carry rater information once its posterior collapse is prevented (KL weight selected as 0 or 0.001; +0.087 over its own unpersonalised version), but only lifts it to the level of the text-only model; PAL-B's few-shot user weights give no gain. Their user representations are unrelated to response style (cross-validated $R^2 \approx 0$ against the style offsets). On binary labels — the setting in which both methods were proposed — every method's personalisation gain is at most 0.002 nats (small but nonzero for PAL-B with K = 4, and for VPL and the style model at k = 50), and VPL's text head is the best binary model (the style model is 0.010 nats behind it and 0.005 behind the text-only model, differences of the size of item-sampling noise): the individual differences that are predictable for new raters live in the graded levels and ties that binary preference data discard. VPL's per-response reward head is slightly better at predicting *which* response is preferred (0.677 vs 0.668), an effect present without personalisation and not significant once prompt groups are also resampled (style model − VPL: −0.9 points [−2.7, +0.8]). The offsets are also a plug-in: added to VPL (fitted on the same k ratings, VPL frozen) they improve it by +0.088 [+0.066, +0.111] nats per rating at k = 20, tie with the style model on likelihood (−0.004 [−0.014, +0.005]) and keep VPL's direction accuracy (+1.1 points over the style model alone; rater bootstrap only).

5.6. Summary of baselines (MultiPref, new raters, k = 20)

Gains are relative to each block's text-only model (D, Dp, D-LoRA); the LoRA block uses adaptation draw 0 only (the draw the in-context model was evaluated on), hence the style model's +0.092 here against +0.100 averaged over five draws in §5.4.

Rows within a block share the text model and the raters; across blocks the text models differ (text-only direction accuracy 0.674 / 0.670 / 0.688). VPL's and PAL-B's own gains over their unpersonalised versions are +0.087 and −0.000.

5.7. Could these methods have detected content personalisation? A semi-synthetic positive control

A ranking null is informative only if the methods would detect content-dependent preferences where they exist. We keep MultiPref's structure (raters, items, who rated what, folds and k-shot draws) and replace the labels by semi-synthetic ones. A generator — the text-only head plus scale-use offsets, fitted on all ratings — gives each rating a distribution, to which a rater-specific content term $(\theta_r^\top g_i)\,(-2,-1,0,1,2)$ is added to the logits; g_i are two

text model	personalisation	gain in LL per rating vs the block's text-only
frozen features (joint item embedding)	group offsets (crowd / expert)	≈ 0
	low-rank latent, tuned prior	+0.095 [+0.069, +0.122]
	style offsets, 3 numbers (DFs)	**+0.100 [+0.074, +0.128]**
frozen features (per-response embeddings)	VPL (reimplemented)	+0.002 [−0.008, +0.012]
	PAL-B (reimplemented)	−0.065 (worse)
	DFs	**+0.094 [+0.069, +0.120]**
	VPL + style offsets	+0.090
LoRA-fine-tuned Qwen2.5-7B (209 raters)	in-context rater history (RC-LoRA, mean of 3 training seeds)	+0.051 [+0.024, +0.079]
	DFs	**+0.092 [+0.063, +0.125]**

features of the response pair (the log length ratio and a random direction in the per-response embedding space, orthogonal to the consensus quality direction) and $\theta_r \sim N(0, \tau^2 I)$. The scale τ is calibrated on the generator alone so that an oracle that knows θ_r gains +1.7, +5.2, +10.1 or +21.1 points of direction accuracy on the test ratings; $\tau = 0$ is a null control. Every method is then run under the paper's protocol. Because a method whose unpersonalised state is not the population state can improve with anyone's ratings, the contrast is **the method adapted on the rater's own k ratings minus the same method given another test rater's adapted parameters** (a random derangement within fold, k and draw). Before the runs we fixed that the protocol has power at a strength if some content-interacting method has a lower 95 % bound above +1 point at k = 20; the +10 and +21 panels and the untuned-prior low-rank runs were added after the first results and are exploratory.

Own minus another rater's k = 20 ratings, direction accuracy in points, 95 % rater bootstrap; 209 new raters, 5 folds. The low-rank latent uses the few-shot-tuned prior. Last column: PAL-B on the full 3,584-dimensional features instead of their PCA-256 (exploratory; $\tau 2$ not run).

No method has power at the pre-specified strengths. In the exploratory dose-response, VPL and PAL-B detect content preferences worth +21 points but recover less than half a point of +10; the low-rank latent, whose user term interacts with a joint embedding of the comparison, has a signal at +21 points that stays below the threshold, and with the untuned prior it shows nothing up to +5 points either. On the real data VPL and PAL-B are at the null-control level, and the low-rank latent shows a marginal +0.30 [−0.00, +0.60]. Read together: on MultiPref, content-level heterogeneity strong enough for an oracle to gain about 20 points of direction accuracy is excluded; heterogeneity worth up to about 10 points cannot be seen by these methods from 20 ratings, so the ranking null of §6 says nothing about it. Because the compressed features might be what limits the content-interacting methods, we re-ran VPL and PAL-B on the full features (exploratory). The text-only fits improve (k = 0 log-likelihood: PAL-B −1.533 vs −1.558, VPL −1.495 vs −1.574), but VPL's latent then collapses and is unused at every strength (own − other exactly 0), so this check rests on PAL-B. Its detection limit drops: it passes the threshold at +10 points and, with k = 50, detects +5 points (+1.05 [+0.39, +1.74]; real data +0.19 [−0.14, +0.49]). The real data stay at the null level (+0.11 [−0.16, +0.39] at k = 20). With this stronger detector, content heterogeneity worth about 10 oracle points of direction accuracy is excluded at k = 20, and about 5 points at k = 50 (the intervals overlap only slightly); the compressed features used for the main tables make the published methods weaker detectors than they need to be. A split-half estimate of each rater's transferable content preference (per-rater ridge weights on 32 content components) had no power even at +5 points and is not used.

6. What personalisation buys: calibration, not ranking

Binary labels remove the signal. On MultiPref with ties dropped and labels binarised, every method's personalisation gain is at most 0.002 nats per rating (§5.5), and the scale-use baseline's binary accuracy equals the text-only model's (−0.0002). The individual differences that are predictable for new raters live in the graded levels and in ties — exactly what binary preference data discard, and the format in which most personalised reward models are trained and evaluated. **Ranking moves little.** Text-independent offsets change how strongly a rater is predicted to prefer a response, not which response: on MultiPref the scale-use baseline raises direction accuracy by only +0.005 [+0.001, +0.008] at k = 20 ([−0.001, +0.011] with items also resampled; about half of it, +0.002 [+0.000, +0.005], is already present at k = 0 and comes from training the text head jointly with the offsets) (and the 5-offset version with an A-vs-B term lowers it by 0.014), while RPS drops from 0.1593 to 0.1532. On single-response safety ratings the same offsets do move the safe/unsafe decision (+4.8 points on DICES), because there a rater's threshold *is* the decision. Content-interacting personalisation does not do better on ranking: VPL's advantage in direction accuracy is present without personalisation (k = 0), comes from its text head, and is within item-sampling noise. How strong a content preference would have to be for these methods to detect it is measured in §5.7. Against its own k = 0 version, no method's direction accuracy on MultiPref rises by more

panel	oracle gain	low-rank latent (DL)	VPL	PAL-B	PAL-B, full features
null control ($\tau = 0$)	0	−0.10 [−0.39, +0.15]	−0.04 [−0.08, +0.01]	−0.23 [−0.46, +0.00]	−0.04 [−0.24, +0.16]
injected, $\tau 2$	+1.7	−0.01 [−0.27, +0.25]	+0.01 [−0.07, +0.11]	−0.10 [−0.37, +0.14]	—
injected, $\tau 5$	+5.2	−0.02 [−0.30, +0.26]	+0.01 [+0.00, +0.03]	+0.23 [−0.04, +0.52]	+0.26 [−0.15, +0.67]
injected, $\tau 10$ (exploratory)	+10.1	+0.09 [−0.18, +0.36]	+0.27 [+0.02, +0.55]	+0.42 [−0.12, +0.95]	+1.95 [+1.19, +2.78]
injected, $\tau 20$ (exploratory)	+21.1	+1.47 [+0.82, +2.14]	+6.92 [+5.59, +8.38]	+4.62 [+3.51, +5.80]	+8.13 [+6.50, +9.85]
real MultiPref	—	+0.30 [−0.00, +0.60]	−0.01 [−0.05, +0.02]	−0.43 [−0.72, −0.15]	+0.11 [−0.16, +0.39]

than 0.2 points at k = 20, and every upper 95 % bound stays below +0.8 points even when prompt groups are resampled together with raters (VPL +0.07 [−0.57, +0.74], style model −0.10 [−0.69, +0.54], PAL-B −0.48 [−1.11, +0.14]). **The ranking null replicates on real users' own choices.** PRISM users each saw 2–4 responses to their own prompt, scored them and chose one; pairing the chosen response with every other one gives 41,199 native pairwise choices of 1,396 users. Under the same new-user protocol (5 user folds × 5 conversation folds, k of the user's pairs from training conversations), we fixed before any result that the ranking null would be stated only if every method's improvement in choice accuracy over its own k = 0 version had an upper 95 % bound below one point at k = 20. It holds for all of them (1,033 new users): VPL +0.12 points [−0.08, +0.33], PAL-B +0.0 [−0.4, +0.4] (K = 2) and +0.3 [−0.1, +0.7] (K = 4), the 5-offset model −0.6 [−1.4, +0.2]; the symmetric scale-use model is identical to the text-only model on binary choices, because a binary label carries no intensity. Personalisation also leaves the binary log-likelihood unchanged (VPL −0.0002, PAL-B −0.0002 nats; the 5-offset model loses 0.003). Without personalisation (k = 0), PAL-B's text head is 1.7 points less accurate than the text-only model and VPL's 0.4 points more. With a graded label cut from the users' own score differences (thresholds fixed in advance), the MultiPref pattern returns at half the size: at k = 20 the scale-use model gains +0.043 [+0.031, +0.057] nats per rating over its k = 0 version, VPL +0.015 [+0.005, +0.027], PAL-B 0.000; the scale-use model is ahead of VPL by +0.045 [+0.030, +0.062]; VPL plus the offsets gains +0.060 and ties the scale-use model (−1.4360 vs −1.4354); and no method's own direction-accuracy gain exceeds +0.2 points (the 5-offset model loses 1.7). **Scale use does not corrupt the shared reward model.** Training the text model jointly with rater offsets and then using the population-mean offsets (k = 0) does not give a better consensus model than ordinary training: −0.003 [−0.010, +0.003] nats on MultiPref (direction +0.3 points, n.s.), −0.022 [−0.037, −0.008] on DICES, +0.001 on PRISM, −0.009 [−0.016, −0.004] with the LoRA reward model. Ordinary training averages scale use out; it matters for predicting a person, not for learning the content preference.

7. Recommendations

(i) Collect and keep graded labels (and ties) when individual differences matter; binary preferences erase most of what is predictable about a new rater. (ii) Report a scale-use calibration baseline (a few text-independent offsets per rater) next to any personalised reward model. Beating it on likelihood is necessary but not sufficient evidence of content personalisation; a method that learns content preferences should show it in ranking or choice outcomes, where the offsets cannot help by construction. (iii) Evaluate on raters unseen in training, adapted from k of their own ratings, and on held-out items, and compare each method with itself given another rater's ratings (a permutation control): a poor unpersonalised state inflates gains measured against k = 0. (iv) Report calibration (log-likelihood, RPS) and ranking (direction accuracy) separately: they answer different questions — how strongly someone prefers, and what to show them. (v) Check what a learned user representation encodes; in the two published models we reimplemented it was unrelated to scale use ($R^2 \approx 0$) and carried little else that transferred to new raters.

8. Discussion

Why did published personalised models learn little about new raters here? Their evaluations differ from ours in three ways that each favour content-dependent personalisation: synthetic users whose preferences differ in content by construction, binary labels that remove scale use, and users seen during training. On real raters, graded labels and unseen users, the structure that transfers from k ratings to new items is mostly how the rater uses the scale, which neither a content-interacting latent (VPL) nor a mixture of content prototypes (PAL-B) represents ($R^2 \approx 0$ against the offsets). Adding the three offsets to VPL recovers the whole gain (+0.088), so the two kinds of personalisation are complementary rather than competing; a method that also captures content-dependent individual preferences should be shown to beat VPL plus offsets, not the text-only model. The positive control (§5.7) adds a second reason: on the compressed features these methods do not recover content preferences worth up to about 10 points of direction accuracy to an oracle from 20 ratings; on full features PAL-B does, and still finds nothing on the real data, which excludes heterogeneity

of that size but not smaller. **What this does and does not say about pluralism.** Our results do not show that people agree about content: the item-level disagreement that replicates across rater panels is real (split-half reliability 0.25 on MultiPref) and about a tenth of it is decodable from text. They show that, for predicting an individual new rater from few ratings, that content-level disagreement is a small part of what is predictable, and that demographic or crowd/expert groups add nothing beyond the text. A reward model that serves different groups different content needs evidence that goes beyond likelihood gains on graded ratings — ranking or choice outcomes for those groups — because scale use alone can produce large likelihood gains without changing which response anyone prefers. **Measurement theory as a baseline generator.** Rater severity and scale use are standard facets in many-facet Rasch measurement and in survey methodology. Their transfer to reward modelling costs three parameters per rater and a convex fit, and it should be the first baseline any personalised reward model is compared against.

9. Protocols and what did not hold

All analyses are documented in an analysis log: timestamped internal protocols written before each stage (not public registrations, so they document rather than certify the order of decisions); changes after results are logged as amendments with reasons, and the symmetric style model DFs and its hypothesis were added after the first full result. Not supported or not robust: (i) the stage-1 decision rule returned "mixed" (text decodability not ≤ 10 %); (ii) non-inferiority of the 5-parameter DF to the low-rank latent (DF $-$ DL ≥ -0.005) holds with the training-rater prior but not with a few-shot-tuned prior on MultiPref (-0.015) and is formally not met on PRISM (bound -0.006); (iii) "scale beats shift" holds only on MultiPref; (iv) the originally registered probit-family hypothesis is not supported (F1 $-$ D = +0.020 [-0.001, +0.041], the primary probit contrast after Amendment 1; homoscedastic F $-$ D = -0.029 [-0.049, -0.008]; superseded when the primary family was moved to the distributional head); (v) the prediction that published user latents encode style failed ($R^2 \approx 0$); (vi) the hybrid's non-inferiority to DFs is formally not met (bound -0.014). Four baseline problems found during the study were fixed in favour of the baselines — a missing constant column in the low-rank latent, a scale non-identifiability in the probit group model, VPL's posterior collapse, and PAL-B's near-unregularised adaptation — and the runs before each fix are reported.

10. Limitations

Frozen-feature text heads for the main tables (a fine-tuned head in §5.4); VPL's latent collapses on full features, so the full-feature bound rests on one method (PAL-B); the positive control injects two kinds of content preference (response length and one embedding direction) on MultiPref only, and other kinds may be easier or harder to detect; three datasets, one language; PRISM scores binned to 5 levels; rater offsets are text-independent by design — interactions with content exist in principle and are what the low-rank latent can capture; we test linear decoders for the text-side measurement only; adaptation uses the rater's ratings of other items, not of the same item.

A. Protocols and amendments

All analyses were specified in timestamped internal protocols (PROTOCOL_STAGE1, PROTOCOL_STAGE2, PROTOCOL_STAGE2B, PROTOCOL_STAGE2C, PROTOCOL_STAGE3); stages 2b and 2c were written after the stage-2a results. Changes after a result were logged as numbered amendments with the reason; the main ones: the first decoding split (a = b = 2) was structurally empty (two ratings determine their distribution) and was replaced by a = 3, b = 1; hand-set penalties in the rater–item model over-fitted and were replaced by inner cross-validation (which reversed an earlier, wrong conclusion that rater shifts do not help); penalty grids were extended when selections fell on grid edges, before any hypothesis was read; the low-rank baseline lacked a constant column and was fixed in its favour; the probit group model had a scale non-identifiability and was fixed; VPL suffered posterior collapse and its KL weight was re-selected including 0; PAL-B's adaptation ridge was re-selected by k-shot validation. Every superseded full run is kept and reported in the supplementary results; two debugging smoke runs and one fold whose output was deleted unread when the penalty grids were extended are not.

B. Hypotheses and outcomes

Stage 2a (MultiPref, k = 20): H1 DF $>$ D — holds (+0.080); H5 (probit F1 $>$ D, the primary probit contrast after Amendment 1) — not supported (+0.020 [-0.001, +0.041]; homoscedastic F $-$ D -0.029), superseded by H1 in Amendment 2; H2 DF $>$ group — holds; H3 DF $\geq$ low-rank $-$ 0.005 — holds under the training-rater prior, not robust under a few-shot-tuned prior (-0.015); H4 rater scale $>$ rater shift — holds on MultiPref only; H6 symmetric style model gains likelihood without losing direction accuracy — holds and robust. Replications: H1r, H2r hold on DICES and PRISM; H3r holds on DICES (robust) and is a tie on PRISM under the tuned prior. Stage 2b: B1, B2, B3 hold. Stage 2c: C1 holds (graded), C2 holds (binary: |DFs $-$ D| accuracy -0.0002; all personalisation gains ≤ 0.002 nats), C3 fails (published latents do not encode style, $R^2 \approx 0$), C4 not met as a conjunction (first part holds, +0.088 over VPL; second part a tie with DFs, lower bound -0.014); Stage

3 (positive control): $\tau 0$ control passes; power at the pre-specified strengths (+1.7, +5.2 oracle points) not established — no content-interacting method's own − other gain has a lower bound above +1 point; exploratory dose-response: VPL and PAL-B pass at +21 points, none at +10; the split-half content estimator (S3-C) has no power; with full features (exploratory, Amendment 5) the $\tau 0$ control passes and PAL-B passes at +10 points (k = 20), so the PCA bottleneck explains part of the detection limit. R1 (PRISM native pairwise choices: every method's own choice-accuracy gain at k = 20 has an upper bound $<$ +0.01) holds (largest bound +0.0071, PAL-B K = 4).

C. Implementation

Text features: last-token final-layer states of Qwen2.5-7B-Instruct under a 5-point judge prompt (MultiPref, PRISM) or a 3-level safety prompt (DICES), label-free PCA to 256 dimensions. Folds: 5 rater folds (stratified by group) $\times$ 5 prompt-group folds; validation: 15 % of training prompt groups. k-shot draws: 5 per k, identical across all models (RC-LoRA: draw 0 only). PRISM item folds are by conversation. Bootstrap: rater clusters, B = 2,000. LoRA: r = 16 on all attention and MLP projections, lr 10^{-4}, one epoch, accumulation 8. VPL / PAL-B: official architectures (latent 512, hidden 512; d_pref 512, mlp2-gelu projectors, angle learner) with an ordered-logistic link for graded labels; VPL's KL weight selected from $\{0.01$ (official), $0.001, 0\}$ and PAL-B's adaptation ridge by k-shot validation (non-official choices made to prevent collapse and over-fitting).

References

Aroyo, L., Taylor, A. S., Díaz, M., Homan, C. M., Parrish, A., Serapio-García, G., Prabhakaran, V., and Wang, D. DICES dataset: Diversity in conversational AI evaluation for safety. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023. doi: 10.52202/075280-2321.

Baumgartner, H. and Steenkamp, J.-B. E. M. Response styles in marketing research: A cross-national investigation. *Journal of Marketing Research*, 38(2):143–156, 2001. doi: 10.1509/jmkr.38.2.143.18840.

Bose, A., Xiong, Z., Chi, Y., Du, S. S., Xiao, L., and Fazel, M. LoRe: Personalizing LLMs via low-rank reward modeling, 2025. URL https://arxiv.org/abs/2504.14439.

Chen, D., Chen, Y., Rege, A., Wang, Z., and Vinayak, R. K. PAL: Sample-efficient personalized reward modeling for pluralistic alignment. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

Dawid, A. P. and Skene, A. M. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 28(1):20–28, 1979.

Deng, N., Zhang, X. F., Liu, S., Wu, W., Wang, L., and Mihalcea, R. You are what you annotate: Towards better models through annotator representations. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 12475–12498, 2023.

Dong, Y. R., Hu, T., Liu, Y., Üstün, A., and Collier, N. When personalization meets reality: A multi-faceted analysis of personalized preference learning, 2025. URL https://arxiv.org/abs/2502.19158.

Eckes, T. *Introduction to Many-Facet Rasch Measurement: Analyzing and Evaluating Rater-Mediated Assessments*, volume 22 of *Language Testing and Evaluation*. Peter Lang, Frankfurt am Main, 2011. ISBN 978-3-631-61350-4.

Gordon, M. L., Lam, M. S., Park, J. S., Patel, K., Hancock, J. T., Hashimoto, T., and Bernstein, M. S. Jury learning: Integrating dissenting voices into machine learning models. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, 2022. doi: 10.1145/3491102.3502004.

Hovy, D., Berg-Kirkpatrick, T., Vaswani, A., and Hovy, E. Learning whom to trust with MACE. In *Proceedings of NAACL-HLT*, pp. 1120–1130, 2013.

Kim, G. and Kim, E. Swap-guided preference learning for personalized reinforcement learning from human feedback. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://arxiv.org/abs/2603.12595.

Kirk, H. R., Whitefield, A., Röttger, P., Bean, A. M., Margatina, K., Ciro, J., Mosquera, R., Bartolo, M., Williams, A., He, H., Vidgen, B., and Hale, S. A. The PRISM alignment dataset: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Datasets and Benchmarks Track*, 2024. doi: 10.52202/079017-3342.

Koren, Y. and Sill, J. OrdRec: An ordinal model for predicting personalized item rating distributions. In *Proceedings of the Fifth ACM Conference on Recommender Systems (RecSys)*, pp. 117–124, 2011.

Koren, Y., Bell, R., and Volinsky, C. Matrix factorization techniques for recommender systems. *Computer*, 42(8): 30–37, 2009.

Li, X., Zhou, R., Lipton, Z. C., and Leqi, L. Personalized language modeling from personalized human feedback, 2024.

Linacre, J. M. *Many-Facet Rasch Measurement*. MESA Press, Chicago, 1989.

Miranda, L. J. V., Wang, Y., Elazar, Y., Kumar, S., Pyatkin, V., Brahman, F., Smith, N. A., Hajishirzi, H., and Dasigi, P. Hybrid preferences: Learning to route instances for human vs. AI feedback. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 7162–7200, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.355.

Mishra, P., Rastogi, C., Pfohl, S. R., Parrish, A., Teh, T. H., Patel, R., Diaz, M., Wang, D., Paganini, M., Prabhakaran, V., Aroyo, L., and Rieser, V. Decoding safety feedback from diverse raters: A data-driven lens on responsiveness to severity. *Transactions on Machine Learning Research*, 2026. URL https://arxiv.org/abs/2503.05609.

Mostafazadeh Davani, A., Díaz, M., and Prabhakaran, V. Dealing with disagreements: Looking beyond the majority vote in subjective annotations. *Transactions of the Association for Computational Linguistics*, 10:92–110, 2022. doi: 10.1162/tacl_a_00449.

Orlikowski, M., Röttger, P., Cimiano, P., and Hovy, D. The ecological fallacy in annotation: Modelling human label variation goes beyond sociodemographics. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 1017–1029, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-short.88.

Plank, B. The "problem" of human label variation: On ground truth in data, modeling and evaluation. In *Proceedings of EMNLP*, pp. 10671–10682, 2022.

Poddar, S., Wan, Y., Ivison, H., Gupta, A., and Jaques, N. Personalizing reinforcement learning from human feedback with variational preference learning. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. doi: 10.52202/079017-1664.

Raykar, V. C., Yu, S., Zhao, L. H., Valadez, G. H., Florin, C., Bogoni, L., and Moy, L. Learning from crowds. *Journal of Machine Learning Research*, 11:1297–1322, 2010.

Ryan, M. J., Shaikh, O., Bhagirath, A., Frees, D., Held, W., and Yang, D. SynthesizeMe! inducing persona-guided prompts for personalized reward models in LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. URL https://arxiv.org/abs/2506.05598.

Siththaranjan, A., Laidlaw, C., and Hadfield-Menell, D. Distributional preference learning: Understanding and accounting for hidden context in RLHF. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2312.08358.

Sorensen, T., Moore, J., Fisher, J., Gordon, M. L., Mireshghallah, N., Rytting, C. M., Ye, A., Jiang, L., Lu, X., Dziri, N., Althoff, T., and Choi, Y. Position: A roadmap to pluralistic alignment. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 46280–46302, 2024.

Uma, A. N., Fornaciari, T., Hovy, D., Paun, S., Plank, B., and Poesio, M. Learning from disagreement: A survey. *Journal of Artificial Intelligence Research*, 72:1385–1470, 2021.

Zhang, M. J. Q., Wang, Z., Hwang, J. D., Dong, Y., Delalleau, O., Choi, Y., Choi, E., Ren, X., and Pyatkin, V. Diverging preferences: When do annotators disagree and do models know? In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 76193–76212, 2025.